A Mincerian Argument for Vocational Education

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Abstract

This investigation uses computational simulation, with a modified Mincer equation to compare an educational model which allows individuals to split years of education between vocational and academic studies, to one which exclusively allows individuals to pursue academic education. While holding average output constant between models, in the vocational and academic studies model, inequality, as measured by the Gini coefficient, is shown to reduce by 29.5%, and poverty is shown to decrease by 37.6%, 62.0%, and 75.2% across FGT_0, FGT_1, and FGT_2 metrics respectively, as compared to the academic-only educational model. Universal academic education, another educational model meant to address inequality and poverty, shows measures of inequality which are 28.1% lower, a poverty headcount index which is 75.1% lower, and a poverty gap which is 29.4% lower than those under the Vocational/Academic Mix model, however it achieves these results with 8.5% lower economic output, and with an FGT_2 metric 51.1% higher than in the Vocational/Academic Mix model. These findings suggest that, compared to the Universal Academic Education model, the Vocational/Academic Mix model is better able to avert the most extreme forms of poverty, while avoiding any decrease in total output. This investigation demonstrates that high skill covariance and low valuation of vocational education increase metrics of poverty and inequality, and, in the case of undervaluation of vocational education, decrease average economic output. However, such effects are modest, and, in most circumstances, the Vocational/Academic Mix model outperforms the academic education-only model along relevant poverty and inequality metrics.

Introduction

Vocational education has often been praised for its ability to promote economic equality (Aizenman, 2017; Ihuoma and Ani, 2024), reduce poverty (Hilal, 2012; Ihuoma and Ani, 2024), and foster economic growth (Aizenman, 2017; Hilal, 2012; Nilsson, 2010). Target 4.3 of the UN Sustainable Development Goals calls for countries, "by 2030, [to] ensure equal access for all women and men to affordable and quality technical, vocational and tertiary education, including university" (UN, 2015). Similarly, the European Union, through the Osnabrück Declaration (2020), acknowledges vocational education's ability to ensure a "fair distribution of the benefits of growth among all citizens and societies," and calls for member countries to "develop and strengthen centres of vocational excellence" (European Union, 2020).

This investigation supports these assertions of benefits of vocational education, and shows that, under an expansion of vocational education, we can expect a reduction in inequality and poverty, even if we assume no change in total output. This investigation finds that, as an economy transitions from one with only academic education, to one with both academic and vocational education, inequality, as measured by the Gini coefficient, falls 29.5%, and poverty, as measured by the poverty headcount ratio, is reduced by 37.6%. Even more encouragingly, after an educational system transitions to including vocational education, the most extreme forms of poverty are those which are most curtailed. This investigation finds that the squared poverty gap is reduced by 75.2% after switching to a vocational-inclusive education model.

In pursuit of similar goals of limiting inequality and poverty, many individuals advocate for universal education (Azarnert, 2010; Wright and Hugo, 2017); SDG Target 4.1 aims to "by 2030, ensure that all girls and boys complete free, equitable and quality primary and secondary education leading to relevant and effective learning outcomes" (UN, 2015). Under a model of universal education, costs of education are externalized (typically born by the state), and individuals are able to (or often required to) continue schooling such that their personal income is enabled to grow, oftentimes past what would be optimal if costs were internalized. Throughout this investigation, the proposed model of a mix of vocational and academic education will be compared to this alternative model of universal education. While this investigation does find that universal education achieves goals of limiting poverty and inequality, it does so at the expense of total output, while failing to limit the most extreme forms of poverty.

In much of the supporting literature, the stated benefits of vocational education of greater economic equality and poverty reduction are attributed to vocational education's low cost (Schuss, 2023), and its subject matter focus of teaching skills which are demanded by the job market (Nilsson, 2010; Sillman and Virtanen, 2022). This paper theorizes that, even without those conditions, and even if one assumes total output is held constant as vocational education is expanded, one can achieve a reduction in economic inequality and poverty with an expansion of vocational education. Vocational education, this investigation asserts, is able to achieve these goals, not as a result of aforementioned low costs or job-relevance of subject matter, but because (a) it is complimentary to academic education, and (b) it builds upon skills which are weakly covariate with academic education.

Regarding complementarity between vocational and academic education, this investigation asserts that, as individuals use skills from their education to earn incomes, literacy, numeracy, and other learned academic skills are used in conjunction with technical skills learned through their vocational education, such that, for all individuals with a basic level of aptitude in both academic and vocational skills, there is an economic incentive to pursue a mix of vocational and academic

education. This assertion is supported by the expansion and success of "dual study programs", especially in Germany, where educational programs explicitly aim to teach a mix of both vocational and academic subject matter (Ertl, 2020; Hofmann et al, 2023). This complementarity between vocational and academic education would differ from non-complimentary skill sets, such as sports education and academic education. In such a scenario, individuals are incentivized to fully commit to pursuing one type of education, as one's output in a career resulting from one type of education would not benefit from additional years of education along the other career path. This complementarity between vocational and academic learning is what allows for average outcomes across the population to remain constant when transitioning from an academic-only education model to one which enables individuals to split their time between academic and vocational education.

Regarding covariance between abilities, this investigation asserts that the innate abilities which one builds upon to translate years of education into earnings are randomly assigned, such that one's vocational ability is not a strong determinant of his or her academic ability. This differs from, for example, math and science education, where one might expect one's math ability to be high if his or her science ability is high, and vice versa. Ultimately, this weak covariance between academic and vocational abilities is what enables those with poor academic abilities (as in, individuals who are unable to translate years of academic education into economic output) to rely on their vocational abilities to achieve economic output. This property of vocational education, this investigation posits, is the engine of poverty reduction and economic equality; those who were left to achieve low levels of output under an academic-only education model are now able to get a second chance at economic success by pursuing vocational education. Despite the assertion of the importance of low covariance between academic and vocational abilities, findings of greater economic equality and lower poverty continue to hold if one relaxes such an assumption; however, as academic and vocational abilities become more covariate, inequality and poverty approach levels achieved under models where only academic education is offered.

Lastly, one common criticism of vocational education is that, even if vocational education confers economic benefits for those who attend, individuals either underestimate its economic benefits (Hong et all, 2022), or face social costs of attendance (Dräger and Wicht, 2021; McCormick et al, 2019), such that individuals might consistently under-attend vocational education. Despite this possibility, this investigation finds that, due to the asserted decreasing marginal benefits of vocational education, even if individuals choose to pursue fewer years of vocational education than would be economically optimal, they nonetheless attend the first years, which confer the highest returns. As a result, for example, in response to a 50% reduction in vocational education attendance, average economic output across the population is only found to decrease by 4.3%, while measures of poverty and inequality continue to remain far lower than those achieved under a model which does not include vocational education.

Methods

The Mincer equation is one of the most commonly used models for estimating one's earnings as a function of his or her years of education, as well as his or her years of work experience. In its most commonly used form, no explicit cost of education is considered aside from the opportunity cost of foregone years of work, however, in order to isolate the returns to education, a simplified Mincer equation as in formula (1), below, is used.

(1) $$y = \sqrt{s\theta} - sc$$

In this equation, one's output is linked one's innate academic ability (θ), and one's years of schooling (s). With decreasing marginal returns to education alongside a fixed marginal cost of education (defined as c), under this model, each individual continues his or her schooling until marginal costs equal marginal benefits in order to optimize output.

(2) $$s_{optimal} = \frac{\theta}{4c^2}$$

(3) $$y_{optimal} = \sqrt{s_{optimal}\theta} - s_{optimal}c = \frac{\theta}{4c}$$

As our Mincerian production function demonstrates, education serves as an exacerbator of economic inequality: while an individual's output, y, at a low universal level of schooling, is correlated with the square root of one's ability, θ, in a scenario where each individual is allowed to continue education until his or her returns are maximized, such output grows linearly with θ.

While formula (1) may be conducive to estimating returns to education under a model where only one type of education is available, an expanded equation is necessary to understand the additional effects of vocational education. To be conservative, this investigation estimates that attending one year of education under the vocationally-inclusive model only provides economic returns equal to those of half of a year of education under the model outlined in formula (1); however, lower returns to academic education are offset by including a vocational education term with a similar functional form. Conceptually, this tradeoff between economic returns to academic and vocational education represents how two educational systems with equal resources could choose two different models: one which invests completely in academic education and achieves the output function outlined in formula (1), and one which splits its total investment into an academic portion and a vocational portion, achieving the output function outlined in formula (4).

(4) $$y = \sqrt{\frac{1}{2}s\theta} + \sqrt{\frac{1}{2}v\phi} - (s+v)c$$

Here, v and ϕ signify an individual's years of vocational education and vocational ability respectively, serving as the vocational counterparts to the academic s and θ terms.

(5) $$s_{optimal} = \frac{\theta}{8c^2}$$

(6) $$v_{optimal} = \frac{\phi}{8c^2}$$

$$y_{optimal} = \frac{\theta + \phi}{8c} \quad (7)$$

Note that, if we assume equal expectation values for θ and ϕ (meaning our population is, on net, no more able vocationally than academically), the expectation value for y across the two presented models, as outlined in formulas (3) and (7), are equal.

Additional assumptions include that, for future computational simulations, θ and ϕ are normally distributed, each with a mean of 32 and a standard deviation of 16. θ and ϕ will be independently assigned, and let c equal 1. In such a case, the average student under the academic Mincerian model, following formula (1), will achieve 8 years of schooling and the standard deviation for the population will be 4 years of schooling.

Now, allow for three different simulated models. In the first model for universal schooling, individuals do not internalize the costs of schooling (as in, s is determined exogenously) and academic education is determined to be 8 years for all individuals; this model will herein be referred to as Universal Academic Education. Unless stated otherwise, assume, under the Universal Academic Education model, that all figures and statistics represent the model before costs of education are considered. In a second model, allow for individuals to determine their years of education according to their innate academic abilities, θ, in accordance with formula (2), allowing individuals to achieve levels of output according to formula (3); this model will herein be referred to as Mincerian Academic Education. In the final model, permit individuals to split time between academic and vocational education, with years of academic and vocational education determined by formulas (5) and (6) respectively, with output determined by formula (7); this model will herein be referred to as Vocational/Academic Mix.

In each model, 25,000 individuals were simulated, according to the model specifications outlined above. Additionally, as negative years of education are not permitted, individuals with θ or ϕ scores below zero (i.e. more than two standard deviations below the mean) are assigned 0 for s or v respectively.

Results

Following the Lorenz curves from Figure 1, the Mincerian Academic Education model, when compared to either the Universal Academic Education, or the Vocational/Academic Mix model, is demonstrated to achieve high inequality of output. Secondly, as is shown in Table 1, while comparing the Mincerian Academic Education model to the Universal Academic Education model demonstrates a trade-off between output and economic equality, the Vocational/Academic Mix model seems to forego this trade-off by achieving a reduction in inequality with no reduction in average output (as compared to the Mincerian Academic Education model). While advocates for Universal Academic Education may argue that a reduction in inequality is worth the lower output achieved, the Vocational/Academic Mix model is able to achieve a reduction in inequality 71.9% as large as that under the Universal Academic Education model, without any penalty to output.

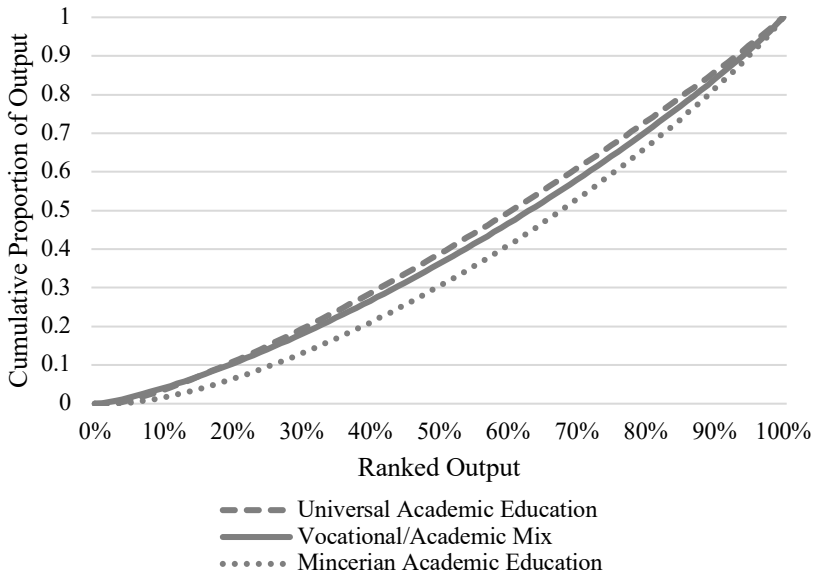


Figure 1. Lorenz Curves of Differing Educational Models

It should be noted that, while the expectation value of y in formulas (1) and (4) should both equal $\theta_{mean}/4c$ in the case that θ and ϕ have equal distributions, average outputs slightly higher than 8 are a result of individuals with negative θ or ϕ scores respectively being assigned s and v values of 0. While formulas (2), (5), and (6) would suggest that individuals with negative θ and ϕ scores would achieve negative years of academic or vocational education, this model does not allow for such.

	Gini Coefficient	Average Output
Universal Academic Education	0.163	7.35[1]
Mincerian Academic Education	0.276	8.04
Vocational/Academic Mix	0.195	8.04

Table 1. Inequality and Output

Poverty Reduction through Vocational Education

It deserves noting that, in Figure 1, the Universal Academic Education curve does not dominate the Vocational/Academic Mix curve for all levels of output. As θ and ϕ are independently assigned, it is individuals with extremely low values of θ who stand to benefit the most from additionally

[1] The average outcome statistic in Table 1 subtracts out the costs of educating each individual ($8c$) from the average output achieved before costs are covered.

including the vocational skills term ϕ in the output function. With this dominance over the Universal Academic Education model at low levels of output, it follows that the individuals with the lowest levels of output in the Vocational/Academic Mix model achieve a larger proportion of total output than the lowest-output individuals in the Universal Academic Education model. The implications for this result on poverty reduction are considerable.

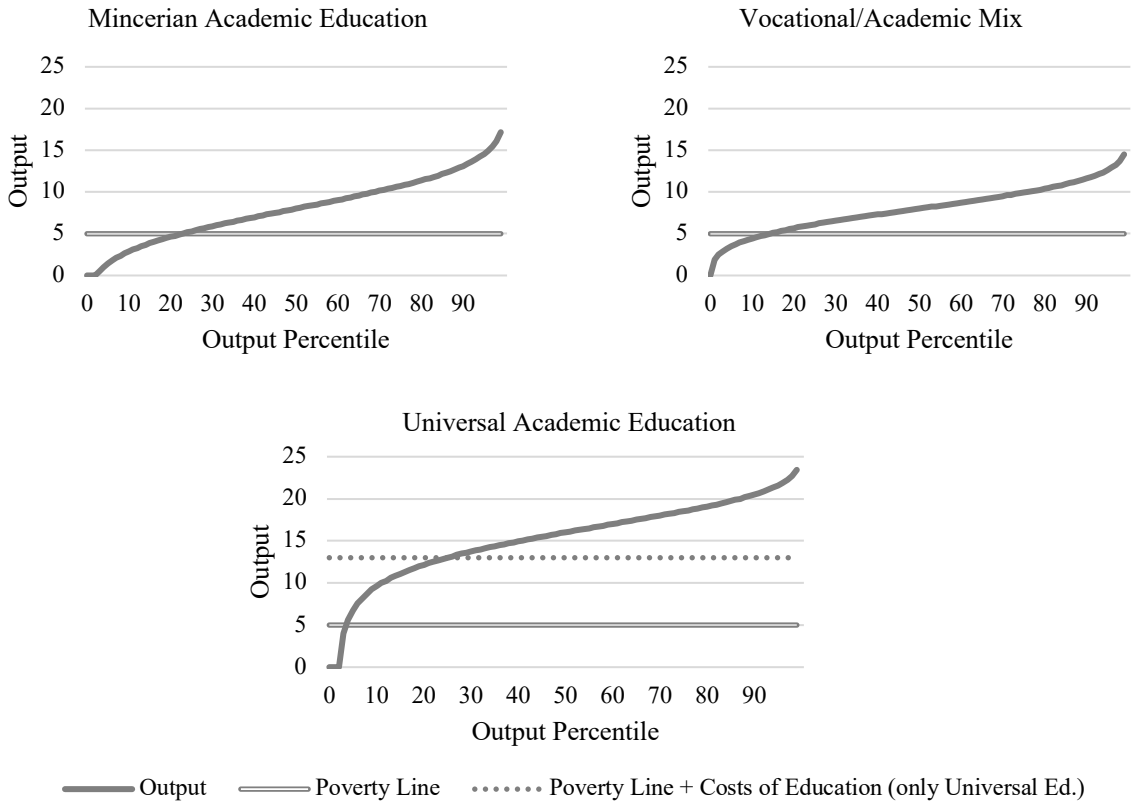


Figure 2. Poverty across Educational Models

Allow for a poverty line to be set equal to 5, defined as the average output of an individual who will complete 5 years of education, according to formula (2). The poverty profiles in Figure 2 demonstrate the levels of output achieved under each model in relation to this poverty line. While the Universal Academic Education poverty profile above crosses the poverty line further to the left than either other model (and therefore achieves a lower headcount poverty ratio than other models), it is important to consider that the figure for Universal Academic Education in Figure 2 is drawn with costs of education unaccounted for.

In the Universal Academic Education model, 3.5% of individuals achieve levels of output below the poverty line; while this is much lower than the 14.2% and 22.8% poverty headcount ratio achieved under the Vocational/Academic Mix and Mincerian Academic models respectively, it remains high in the context of the size of the outstanding costs of education. If, under the Universal Academic Education model, costs of education were covered equally by each individual, the poverty headcount would rise to 24.5%, as represented by the dotted line in Figure 2.

Additionally, the poverty gap is equal to 9.8%, 3.7%, and 2.6% in the Mincerian Academic Education, Vocational/Academic Mix, and Universal Academic Education models respectively. With a poverty line of 5, a 3.7% poverty gap metric suggests the average depth of poverty in monetary terms equals 0.186. Considering the costs of education for each individual in the Universal Academic Education model are equal to 8, such costs equal 43.0 times the average depth of poverty under the Vocational/Academic Mix model, meaning the same funding to provide 8 years of education to each individual in the Universal Academic Education model could fill the poverty gap 43 times over in the Vocational/Academic Mix model. Similarly, in the Mincerian Academic Education model, given and average depth of poverty of 0.490, as conferred from a 9.8% poverty gap, it follows that the costs of education in the Universal Academic Education model are equal to 16.3 times the poverty gap for the Mincerian Academic Education Model.

Addressing the last of the FGT poverty indices, the squared poverty gap index, it becomes increasingly clear how expanding vocational education enables an economy to avoid the most extreme forms of poverty. Across the three models, the squared poverty gap index is equal to 6.5%, 1.6%, and 2.4% in the Mincerian Academic Education, Vocational/Academic Mix, and Universal Academic Education models respectively. As a reminder, in the Universal Academic Education model, costs of education have yet to be addressed, meaning, even if costs are education are entirely foregone, the severity of poverty, according to the squared poverty gap index, is worse in the Universal Academic Education model than it is in the Vocational/Academic Mix model.

Considering a much lower poverty headcount ratio achieved in the Universal Academic Education model, alongside a higher squared poverty gap relative to the Vocational/Academic Mix model, it is clear that, among those who are living in poverty, the depth of poverty is much more severe, on average, in the Universal Academic Education scenario than in the Vocational/Academic Mix model. Specifically, among those living in poverty, the average depth of poverty for individuals in the Universal Academic Education model is 1.19, while the same figure is 0.25 under the Vocational/Academic Mix model.

With differences across models with respect to incidence and depth of poverty, it follows that the relative differences between models along the various poverty metrics is sensitive to the poverty line chosen. Table 2 shows that, across all three poverty metrics, there exist poverty lines for which measures of poverty between the Vocational/Academic Mix and Universal Academic Education models equal.

	Poverty Line	FGT_α Metric
$\alpha = 0$	2.77	2.6%
$\alpha = 1$	4.35	2.5%
$\alpha = 2$	5.90	2.5%

Table 2. Poverty Lines Under Which Universal Academic Education and Vocational/Academic Mix Models Share Equivalent FGT Poverty Metrics

It seems quite interesting that, across FGT poverty metrics, whether they be $\alpha=0$, $\alpha=1$, or $\alpha=2$, the poverty metrics consistently fall close to 2.5% at poverty line specifications set to allow the given poverty metrics to equal across Vocational Academic Mix and Universal Academic Education models. It's not well understood why this is the case, but this deserves further investigation.

Regardless, following from these findings, under poverty line specifications below 2.77, the Vocational/Academic Mix model outperforms the Universal Academic Education model

across all FGT metrics of poverty, while the opposite is true for poverty line specifications above 5.90. Similarly, for all poverty line specifications above average levels of output (under the current model specifications, for any poverty line above 8), FGT_0 is greater under the Vocational/Education Mix model than it is under the Mincerian Academic Education model. Additionally, FGT_1 approaches equality between the Vocational/Education Mix and Mincerian Academic Education models as the poverty line grows to infinity; however, as a result of the relative depths of poverty across the two models, for no positive poverty line will FGT_2 statistics equal between the Vocational/Education Mix and Mincerian Academic Education models.

Considering the ability for the Vocational/Academic Mix model to avoid the most extreme forms of poverty, and the inability for Universal Academic Education to do the same, it follows that, among individuals for whom academic education is unable to make a meaningful difference on an individual's ability to achieve economic output, expanding access to education through cost-covering can only have a moderate effect on output; for such individuals, it's vital that systems of education are able to build upon their skills outside of the academic sphere. It therefore follows that expanded access to vocational education is a powerful tool to reach such left-behind individuals and allow for many to be lifted out of poverty.

Covariance in Ability

Until now, all analysis has assumed θ and ϕ to be independent random draws, that is, one's innate academic ability is not predictive of one's vocational ability. This, however, may not be true. One could imagine possible distributions of skill such that θ and ϕ are highly covariate, meaning one's high θ is a strong predictor of a high ϕ, or highly negatively correlated, such that a high θ is a strong indicator for an individual having a low ϕ.

In order to understand how different covariances between θ and ϕ affect poverty and inequality within the Vocational/Academic Mix model, while maintaining similar distributions of both variables, first, θ scores are generated for each individual from a random normal distribution with a mean of 32 and standard deviation of 16. Second, a separate variable is generated for each individual, equal to $\theta + \varepsilon$, where ε is random normal noise; we can define $\theta + \varepsilon$ as θ'. With θ' determined, a set of ϕ scores are generated, similarly, from a normal distribution with a mean of 32 and a standard deviation of 16. ϕ scores are then assigned to individuals according to their rankings of θ'. For simulations with positive covariance, individuals with the highest θ' are assigned highest ϕ scores, while for simulations with negative covariance, individuals with the highest θ' scores are assigned the lowest ϕ scores. This method of assigning θ and ϕ ensures that both variables maintain normal distributions with means of 32 and standard deviations of 16, while enabling increasing or decreasing covariance by decreasing or increasing the standard deviation of the distribution of ε.

With θ and ϕ defined as each having a standard deviation of 16, following the properties of covariance that cov(X,X) = var(X), and cov(X,-X) = -var(X), we can expect the covariance between θ and ϕ to fall between -256 and 256.

As shown in Figure 3, the inclusion of vocational education is most effective at bridging inequalities in negative covariance scenarios, where θ and ϕ serve as compliments to one another. In these cases, individuals with low levels of θ are more often able to achieve economic output through higher ϕ score and through further years of vocational education, and vice versa.

Covariance in Ability: Inequality

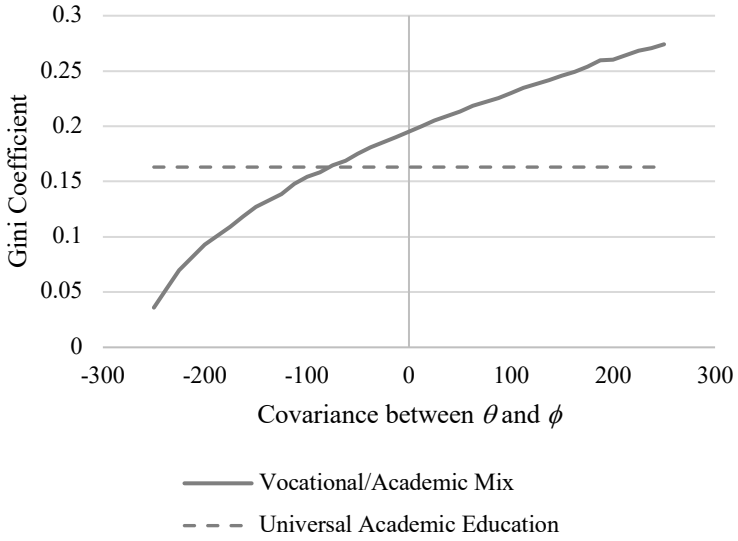


Figure 3. Inequality Across Covariances Between θ and ϕ

The two limits of the trend shown in Figure 3 are also both quite telling. As the covariance between θ and ϕ approaches zero, so does the Gini coefficient, suggesting perfect equality in such a scenario. Below we can rewrite any θ to be equal to $\mu_\theta + \epsilon_\theta$, such that μ_θ is the population mean of θ and ϵ_θ is an individual's deviation from that mean; likewise ϕ can be written as $\mu_\phi + \epsilon_\phi$.

$$y_{optimal} = \frac{\theta + \phi}{8c} = \frac{(\mu_\theta + \epsilon_\theta) + (\mu_\phi + \epsilon_\phi)}{8c} \tag{8}$$

In this case, if ϵ_θ and ϵ_ϕ have equal population-level distributions, the limit of $y_{optimal}$ as the covariance between θ and ϕ approaches its minimum (most highly negative) value, is equal to the limit where ϵ_θ approaches $-\epsilon_\phi$, or otherwise, where $\epsilon_\theta + \epsilon_\phi$ approaches zero.

$$\lim_{cov(\theta,\phi)\to-\sigma_\theta\sigma_\phi} y_{optimal} = \lim_{\epsilon_\theta\to-\epsilon_\phi} y_{optimal} = \frac{\mu_\theta + \mu_\phi}{8c} \tag{9}$$

$$\lim_{cov(\theta,\phi)\to-\sigma_\theta\sigma_\phi} y_{optimal} = \frac{\mu}{4c} \tag{10}$$

In this case, any deviation in academic ability which an individual has from the population mean is zeroed out by an opposite deviation in their vocational ability, such that outcomes do not vary across the population, and all individuals achieve outcomes equal to $(\mu_\theta + \mu_\phi)/8c$. This can be simplified further to $\mu/4c$ if, as in our case, $\mu_\theta = \mu_\phi$, such that we can define a value $\mu = \mu_\theta = \mu_\phi$.

As such, with minimum covariance, equality of outcomes is achieved, and the Gini coefficient equals zero.

In the opposite direction, increasing covariance between θ and ϕ has the effect of increasing the Gini coefficient for outcomes in our population.

$$\lim_{cov(\theta,\phi)\to\sigma_\theta\sigma_\phi} y_{optimal} = \lim_{\epsilon_\theta\to\epsilon_\phi} y_{optimal} = \frac{\mu_\theta + \mu_\phi + 2\epsilon_\theta}{8c} \tag{11}$$

$$\lim_{cov(\theta,\phi)\to\sigma_\theta\sigma_\phi} y_{optimal} = \frac{\mu_\theta + \epsilon_\theta}{4c} = \frac{\theta}{4c} \tag{12}$$

Following again from formula (8), with equal distributions of ϵ_θ and ϵ_ϕ, maximum covariance is achieved if ϵ_θ and ϵ_ϕ are equal for each individual. If, as in our case, $\mu_\theta = \mu_\phi$, the limit of $y_{optimal}$ as the covariance between θ and ϕ approaches its maximum value is equal to the optimal output achieved under the Mincerian Academic Education model, or formula (3). As such, while, as θ and ϕ become more covariate, outcomes across the population become more unequal, any covariance in θ and ϕ below the positive limit allows for more equal outcomes to be achieved in the Vocational/Academic Mix model than under the Mincerian Education model, without any effect on mean output.

Covariance in Ability: Poverty

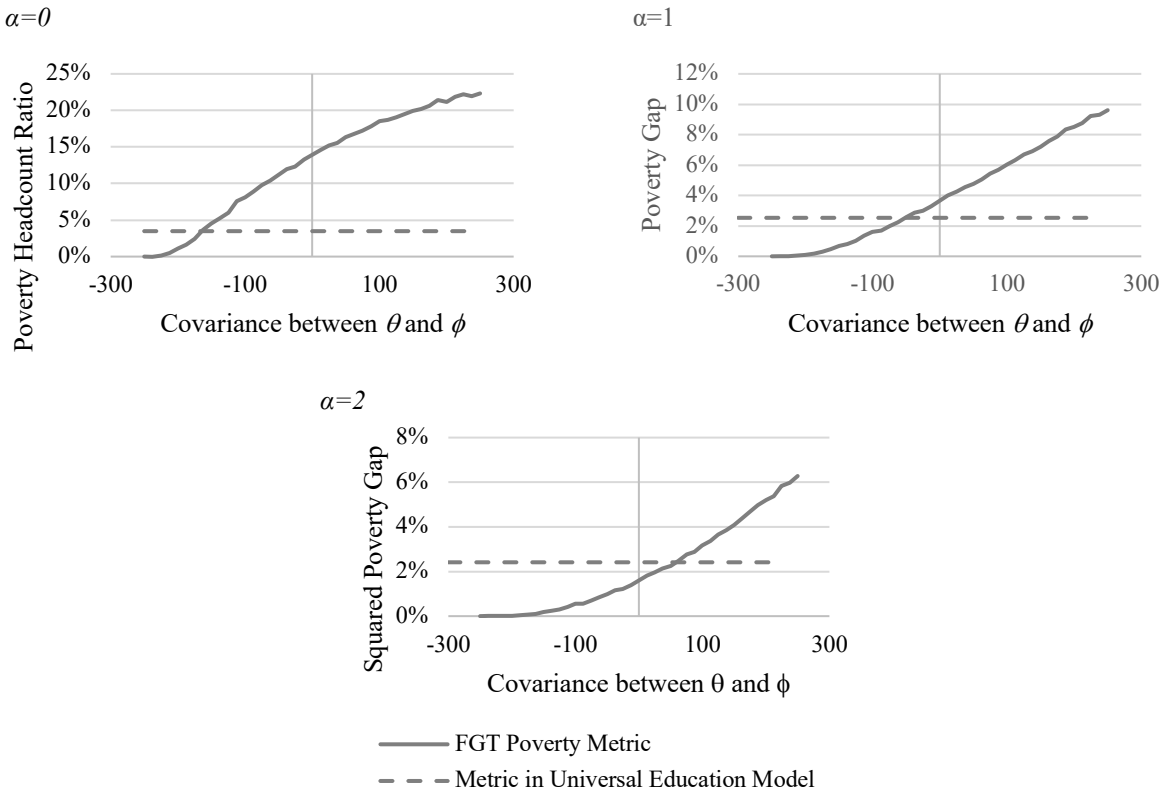


Figure 4: FGT_α Poverty Metrics Across Covariances Between θ and ϕ

Again, as shown in formulas (10) and (12), output is equal to $\mu/4c$ for all individuals in the minimum covariance case, and is equal to $\theta/4c$, under perfect covariance, making it equal to output under the Mincerian Academic Education model. As such, given that $\mu/4c$ is above the poverty line, poverty falls to zero under minimum covariance by all three FGT metrics of poverty. Similarly, as covariance in academic and vocational abilities approach their maximum, FGT metrics of poverty in the Vocational/Academic Mix model approach those of the Mincerian Academic Education model.

While, for any scenario outside of perfect covariance between θ and ϕ, with the model parameters as they are ($\mu_\theta = \mu_\phi = 32$, $\sigma_\theta = \sigma_\phi = 16$, $c = 1$, poverty line = 5) more individuals fall below the poverty line in the Vocational/Academic Mix model than the Universal Academic Education model, the ability for expanded vocational education to avert the most extreme forms of poverty is still true under moderate levels of covariance. With concave curves for FGT_1 and FGT_2 metrics of poverty in Figure 4 above, for small positive deviations in covariance above 0, one can expect a less-than-proportional increases in poverty. This assertion suggests that, outside of large covariation between θ and ϕ, one can expect the finding that the Vocational/Academic Mix model avoids extreme poverty to hold, even as the assumption of independent assignment of θ and ϕ is relaxed.

Undervaluation of Vocational Education

One of the principal critiques laid against vocational education is that individuals underestimate its economic benefits, (Hong et all, 2022) face social costs of attendance, (Dräger and Wicht, 2021; McCormick et al, 2019), or otherwise attend fewer years of vocational education than would be economically optimal. To incorporate this possibility into our model, let us introduce a term for perceived valuation of vocational education, δ, corresponding to the ratio of years one will ultimately spend pursuing vocational education, relative to the economically optimal years of education; as in, if an individual were to have a vocational ability score, ϕ, for which, according to formula (6), he or she would optimally pursue 8 years of vocational education, if the individual undervalues vocational education such that $\delta = 0.75$, this individual would only choose to attend vocational education for 6 years, as demonstrated in formula (13).

$$v = \delta v_{optimal} = \delta \frac{\phi}{8c^2} \tag{13}$$

$$y = \sqrt{\frac{1}{2} s_{optimal} \theta} + \sqrt{\frac{1}{2} \delta v_{optimal} \phi} - \left(s_{optimal} + \delta v_{optimal}\right) c \tag{14}$$

$$y = \frac{\theta}{8c} + \frac{\phi}{8c}\left(2\sqrt{\delta} - \delta\right) \tag{15}$$

With the modifications to the functional form of the production function, it therefore follows that outcomes can be determined using formula (15).

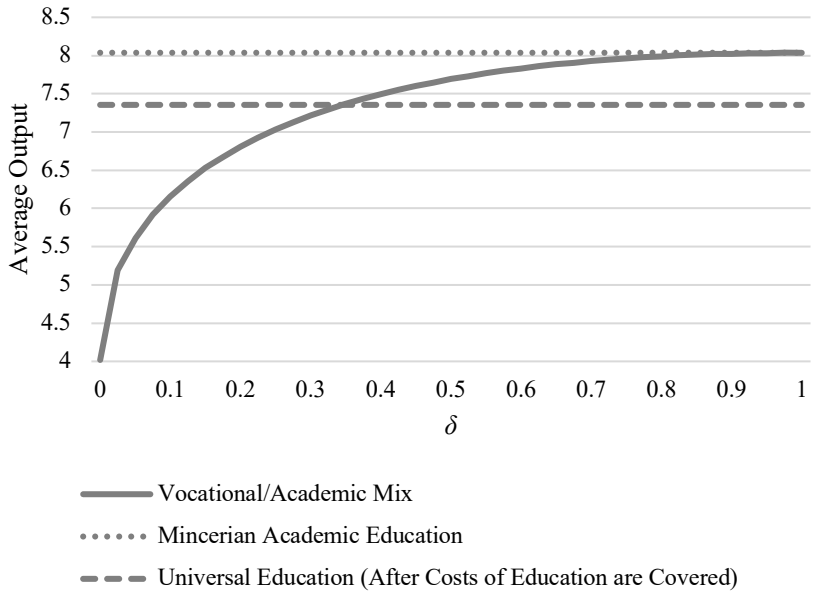


Figure 5: Output under Undervalued Vocational Education

In formula (15), any deviation in perceived valuation of vocational education, δ, from 1, results in less-than-optimal output. As shown in Figure 5, it follows that, as δ approaches 0, the average output achieved across the population approaches half of the output achieved under optimal conditions; with no value placed on vocational education, only returns from academic education are achieved. For many most values of δ, however, output remains relatively close to optimal levels; at $\delta = 0.346$, average levels of output under the Vocational/Academic Mix model equal those of the Universal Academic Education model, after costs of education are covered, equaling $y = 7.35$.

Similar to findings for output, economic inequality, as measured by the Gini coefficient, worsens as δ grows, with its limit at $\delta = 0$ equal to the Gini coefficient under the Mincerian Academic Education model or 0.276. Again, however, similar to findings for output, measures of inequality remain relatively constant under moderate deviations from $\delta = 1$.

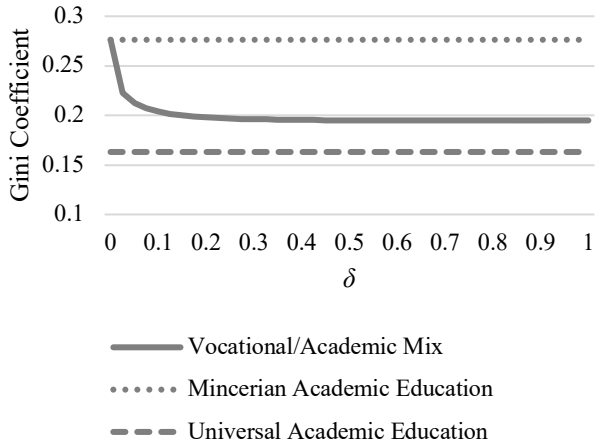


Figure 6: Inequality under Undervalued Vocational Education

Similarly to output and inequality, poverty, across all three FGT measures, worsens as δ decreases. Following the $\alpha = 0$ chart in Figure 7, all individuals who would not be able to achieve outcomes equal to twice the poverty line in the traditional Mincerian Academic Education model would fall into poverty under the Vocational/Academic Mix model at $\delta = 0$. With the parameters used in this model, this would lead to a poverty headcount ratio of 68.7%. Again, however, similar to output and inequality, outside of extremely low values of δ, measures of poverty remain only moderately worsened by a decreasing perceived value of vocational education, δ. For example, FGT_0, FGT_1, and FGT_2 poverty metrics are equal between the Mincerian Academic Education and Vocational/Academic Mix models at $\delta = 0.20$, 0.07, and 0.03 respectively.

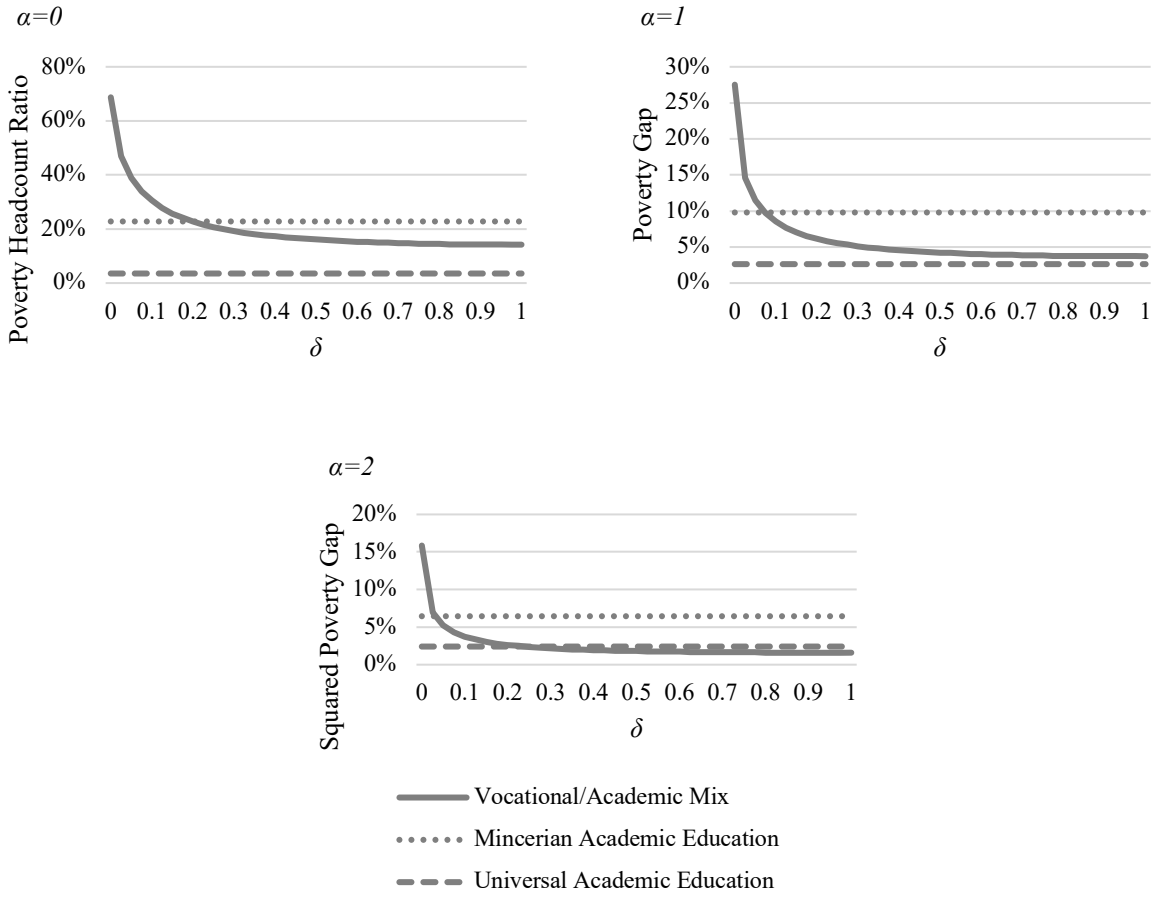


Figure 7: Poverty under Undervalued Vocational Education

As demonstrated throughout, across measures of output, inequality, and poverty, for almost all non-extreme values of δ, changes in relevant metrics away from optimal levels are only moderate. For example, at $\delta = 0.5$, meaning in a scenario where individuals only pursue vocational education for half as many years as would be economically optimal, average output equals 95.7% optimal levels, the Gini coefficient remains only 0.09% above optimal levels, and poverty grows only 13.3%, 13.5%, and 12.9% above optimal levels for FGT_0, FGT_1, and FGT_2 poverty metrics respectively. This is largely a result of decreasing marginal returns to education under the model specified in formula (4), meaning the most valuable years of education are an individual's first

years of education, which are achieved regardless of the poor perceived valuation (low δ) which individuals may have of vocational education.

Additionally, this model for incorporating undervaluation of vocational education assumes that all individuals in the population value vocational education equally, that is, δ is a population-level parameter, and is exogenous of an individual's θ or ϕ. However, if we allow for differentiation in δ across the population, it seems logical that an individual with a high vocational ability (ϕ), and low academic ability (θ) would place higher value on vocational education than other individuals. In such a scenario, it would follow that measures of inequality and poverty would be less affected by undervaluation of vocational education than the analysis above would suggest.

Conclusions

As this investigation theorizes, adapting an education system to allow individuals to split their years of education between academic and vocational studies leads to greater economic equality and lower levels of poverty than are possible under a model which exclusively allows for academic education. Specifically, this investigation suggests that transitioning from an academic-only education model to a vocational/academic mix model leads to a reduction in inequality, as measured by the Gini coefficient, of 29.3%, and a reduction in poverty, as measured by the poverty headcount ratio, of 37.6%. Most encouragingly, the poverty reduction achieved by such a switch between models only grows as measures of poverty put more weight on those facing the most extreme forms of poverty; by switching models, we see a 62.0% and 75.3% reduction in poverty across the FGT_1 and FGT_2 metrics of poverty respectively. As such, not only is poverty reduced by expanding vocational education, but the most extreme forms of poverty are those which are most curtailed.

An alternative model for achieving similar goals of economic equality and poverty reduction is universal academic education, where all individuals achieve equal years of education regardless of their ability. Under this model, while while inequality, as measured by the Gini coefficient, and poverty, as measured by the poverty headcount ratio, are favorable compared to the Vocational/Academic Mix model advocated for in this investigation, it achieves such goals at the expense of economic output (average economic output is 8.6% lower than either the Vocational/Academic Mix or the Mincerian Academic Education model), and while allowing for persistently high levels of extreme poverty (FGT_2 measures of poverty are 33.8% lower under the Vocational/Academic Mix model than the Universal Academic Education model).

While relaxing the assumption that individuals' academic abilities (θ) and vocational abilities (ϕ) are not covariate does infer smaller reductions in economic inequality and poverty than as was true under non-covariate ability scores, only at the limit of perfect collinearity do such statistics approach levels achieved under the traditional Mincerian academic model, meaning even for highly covariate abilities across a population, benefits in inequality and poverty can still be realized by switching from a model with only academic education to one which includes vocational education.

A separate common concern related to vocational education is that individuals undervalue vocational education, either because they underestimate its returns or because they face a social cost of attending which is not captured by economic data, and therefore, individuals attend fewer years of vocational education than would otherwise be economically optimal. While undervaluation of vocational education is a genuine concern, and while implementing potential

undervaluation into the model does reduce average economic output, worsen inequality, and increase levels of poverty, the magnitude of these effects is small for most levels of undervaluation. For example, with a 50% reduction in the years of vocational education achieved below optimal levels, total economic output only sees a 4.3% reduction; inequality, as measured by the Gini coefficient, increases by only 0.09%; and poverty, as measured by FGT_0, FGT_1, and FGT_2, increases by 13.3%, 13.5%, and 12.9% respectively. While these outcomes certainly aren't desirable, with the exception of total output, each of these statistics remains far better than what was achieved under perfect conditions in the Mincerian Academic Education model, suggesting that, even if the population does not value vocational education as highly as would be economically optimal, an educational model which is inclusive of vocational learning will likely still be preferrable to an educational model which excludes vocational education.

Additionally, while the output function for this investigation was chosen to explore changes in inequality and poverty while maintaining the population average of output constant, empirical evidence suggests that expanding vocational education leads to economic growth (Aizenman, 2017; Hilal, 2012; Nilsson, 2010), meaning perhaps the production function used in this investigation underestimates economic output in the vocational/academic mix scenario. In such a case, one would expect similar findings regarding reduced inequality, with even further reductions in poverty, and an increased average output as one transitions from an academic-only education system to one which additionally includes vocational education.

While this model rests on two key assumptions–that vocational and academic ability are not covariate, and that vocational and academic ability are complimentary–only the former was tested during this investigation. When considering further exploration, it would be informative to relax the assumption of complementarity, either by requiring an individual to choose to only pursue vocational or academic education, or otherwise by endogenously penalizing splitting time between academic and vocational education.

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